

Ahmad Hussain, MD

Interventional Radiology Research Fellow at Brigham and Women's Hospital/Harvard Medical School

[Email](#) | [LinkedIn](#) | [ResearchGate](#) | [Portfolio](#)

Education

Bachelor of Medicine, Bachelor of Surgery (MBBS)

2019-25

King Edward Medical University, Pakistan

ECFMG Certified (MD)

2026-28

MyIntealth (USMLE Step 1: **Passed**, USMLE Step 2: **255**, OET: **Passed**, Step 3: **221**)

Bachelor of Science (BSc)

2017-19

University of the Punjab, Pakistan

Research Experience

Postdoctoral Research Fellow

Oct 2025 - Present

Interventional Radiology department at Brigham and Women's Hospital/Harvard Medical School, Boston

PI: Yan Epelboym, MD, MPH

Research Focus: Clinical trials planning and management including patient screening, procedural planning, and angiographic interpretation for MSK embolization (GAE and TAE), with focus on treatment endpoints and longitudinal clinical response.

Optimization of patient selection in knee OA patients, Quantitative angiographic and perfusion imaging, assessing feasibility of MRI instead of Fluoroscopy/CT for embolization, Evaluating feasibility of Microbot (Robot-assisted GAE) in Knee OA patients.

Research Assistant

Nov 2024 - Feb 2025

Biology department (Basic Science Lab) at Lahore University of Management Sciences, Pakistan

PI: Shaper Mirza, PhD

Research Focus: Understanding immune impairments in type-2 diabetes and their impact on pneumococcal infections.

Skills: Bacterial Culturing and Assays, Immunoassays and DNA Sequencing.

Lab Intern

Sep 2024 - Oct 2024

Pathology lab at various locations across Lahore, Pakistan

PI: Sohail Chughtai, MD

Focus: Two months internship program for medical students to develop interest in Pathology with weekly rotations across Histopathology Lab, Tissue Engineering Lab, Genetics Lab and Forensic Lab.

Lab Intern

2019-21

Biochemistry Lab at KEMU, Pakistan

PI: Naqshab Chaudhary, MBBS, PhD

Focus: Tissue Handling, Cell Culture, Flow Cytometry, PCR, RT-PCR, Western Blot, Microscopy, ELISA.

Publications

Published

Selecting Osteoarthritis Patients for Genicular Artery Embolization: The Role of Imaging and Molecular Biomarkers. **Hussain A**, Taslakian B, et al. *Seminars in Interventional Radiology*. 2026. [Full Text Link](#)

Arterial Anatomy of the Shoulder: What the Interventional Radiologist Needs to Know. **Hussain A**, Bihun R, Epelboym Y. *Seminars in Interventional Radiology*. 2026. [Full Text Link](#)

Adhesive Capsulitis Embolization: A Narrative Review. Leung K, **Hussain A** et al. *Skeletal Radiology* 2026 [Full Text Link](#) (Co-first author)

Cystic genitourinary lesions in the pelvis: pearls and pitfalls. Hamade M, et al. *Abdominal Radiology*. 2025;50(12):5963–5983. [Full Text Link](#)

Effectiveness and reliability of AI in diagnosis and robot-assisted spinal and cranial surgery. Shahid I, et al. *Ann Med Surg (Lond)*. 2025;87(11):7236-7243. [Full Text Link](#)

How long to wait after local infiltration anaesthesia: systematic review. Bajwa MS, et al. *BJS Open*. 2023;7(5). [Full Text Link](#)

Redefining Fasciocutaneous Microanatomy: An Illustrated Review of Current Concepts and Their Clinical Correlates. Bajwa MS, et al. International Microsurgery Journal. 2023;7(1):2 [Full Text Link](#)

Impact of trans-anal versus laparoscopic total mesorectal excision on surgical and pathologic outcomes: meta-analysis of randomized controlled trials. Farooqi M, **Hussain A**, et al. Langenbecks Arch Surg. 2023;408(1):413. [Full Text Link](#)

Bevacizumab combination therapy versus standard chemotherapy for ovarian cancer: systematic review and meta-analysis. Rehman OU, et al. Am J Clin Oncol. 2024;47(8):399-408. [Full Text Link](#)

Calcifying aponeurotic fibroma in the deep space of the hand. Antar RM, et al. Plast Reconstr Surg Glob Open. 2024;12(12):e6358. [Full Text Link](#)

An infant with Patau syndrome associated with congenital heart defects. Khan U, **Hussain A** et al. Ann Med Surg (Lond). 2022;80:104100. [Full Text Link](#)

Mixed Trabecular and Mucinous Carcinoid Arising in the Background of a Mature Ovarian Cystic Teratoma. Ali SMH, Khan HA, Malik A, Ali S, **Hussain A**. Cureus. 2025;17(1):e77013. [Full Text Link](#)

Microscopic diascope dermoscopy, an international ultra-budget alternative to traditional dermoscopy. Prideaux A, et al. Clinical and Experimental Dermatology, Feb 2026, [Full Text Link](#)

Conference Abstracts

Hamade M, et al. Cystic genitourinary lesions in the pelvis: pearls and pitfalls. Presented at RSNA Annual Meeting 2024; Chicago, IL.

Nadeem A, et al. Microwave Ablation versus Resection for Colorectal Cancer Liver Metastases: A Systematic Review and Meta-Analysis. SIR Annual Meeting 2025; 36(3):S190-S191.

Submitted/In Press

Hussain A. Artificial Intelligence in Radiology: Beyond Image Interpretation. Journal of the American College of Radiology (JACR).

Accepted for publication, Aug 2026.

Hussain A, et al. Inter- and Intra-reader Reliability of Angiographic Blush Assessment in Genicular Artery Embolization: Native Knee Cohort. Submitted, SIR Annual Scientific Meeting 2027.

Hussain A, et al. Inter- and Intra-reader Reliability of Angiographic Blush Assessment in Genicular Artery Embolization: Total Knee Replacement Cohort. Submitted, SIR Annual Scientific Meeting 2027.

Hussain A, et al. Transarterial Embolization for Symptomatic Hip Osteoarthritis: A Systematic Review and Meta-analysis. Submitted, SIR Annual Scientific Meeting 2027.

Hussain A, et al. Higher Body Mass Index as a Predictor of Clinical Response Following Genicular Artery Embolization for Knee Osteoarthritis: A Systematic Review and Narrative Synthesis. Submitted, SIR Annual Scientific Meeting 2027

Hussain A, et al. Shoulder Artery Embolization for Refractory Glenohumeral Osteoarthritis in a Patient with Active Acute Myeloid Leukemia. Submitted, Society of Interventional Oncology (SIO) IO Essentials (2027)

Clinical Experience

Hands-on Elective (Away IR)

July 2024

Interventional Radiology department at **WashU** in Saint Louis to gain practical experience about advanced IR procedures, including biopsies, vascular access placements, angioplasties, embolization (uterine artery, GI bleeding), sclerotherapy, tumor ablations, abscess drainages, biliary interventions and Y-90 radioembolization for liver cancer.

Clinical Observership

July 2026

University of Maryland Medical Center Radiology Observership to gain exposure to diagnostic radiology practice, with particular emphasis on musculoskeletal radiology, including attending readouts, case discussions, and departmental noon conferences.

Clinical Observership

June 2024- July 2024

Lenox Hill Interventional Radiology to observe about a range of advanced procedures, including Y-90 radioembolization, prostate and uterine artery embolization, fluoroscopic guided biopsies and thrombectomies.

Clinical Rotation (Home DR)

Feb-March 2025

Department of Radiology, Mayo Hospital, Lahore to gain hands-on experience in diagnostic ultrasound and Doppler imaging. Contributed to CT and MRI reporting; performed eFAST in emergency settings. Observed oncologic tumor ablation procedures.

Leadership and Service

Fundraising Manager, Students' Patient Welfare Society, KEMU

Organizer, Monthly Free Eye Camps at Vision of Hope by IFMSA

Co-founder, The Little Steps, an organization to promote personal hygiene in public school kids in Pakistan

Project Director, Volunteer Force Against Polio Transmission (VFAPT)

Student Volunteer, Shaukat Khanum Cancer Hospital Fundraising (Annual Campaign

Student Volunteer, fundraising campaign for flood affectees by Alkhidmat Foundation

Student Volunteer, awareness campaigns for COVID-19 preventive measures and immunization

Student Volunteer, Plant Green (Nationwide plantation drive in Pakistan)

Medical Transcriber and Translator, Sehat Sanjha (App development related to AI in Healthcare at LUMS)

Honors and Awards

Honors in Radiology, Surgery and Internal Medicine Clerkship

GEST Research Trainee Scholarship Award RSNA Trainee Travel Award

Aga Khan University Challenge Scholarship Program

Quaid-e-Azam Scholarship for High Achievers (High School and College)

Gold Medalist at Lahore Table Tennis Intercollegiate Championship 2024

Teaching and Mentorship

Research Mentorship

Mentored medical students about clinical research and particularly about Case Reports, Review Articles, Systematic Reviews and Meta Analyses, helping them learn statistical tools such as Review Manager, MedCalc, and JAMOV. Also, addressed a session on navigating research pathway into US before applying for residency.

USMLE Mentorship

Mentored and tutored 200+ students throughout their USMLE exam journey.

Extracurricular Activities

Table Tennis, Cricket, Soccer, Cycling, Photography and E-Sports

References

To be provided upon request.